

1. To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile.

Task -1

```
vennela_07@VennelaPC:~$ mkdir OSSP
cd OSSP
vennela_07@VennelaPC:~/OSSP$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/vennela_07/OSSP/.git/
vennela_07@VennelaPC:~/OSSP$ mkdir src include obj bin docs tests scripts assets
vennela_07@VennelaPC:~/OSSP$ touch .gitignore README.md LICENSE Makefile
vennela_07@VennelaPC:~/OSSP$ touch src/{main.c,shell.c,parser.c,executor.c,builtin.c}
vennela_07@VennelaPC:~/OSSP$ touch include/{shell.h,parser.h,executor.h,builtin.h}
vennela_07@VennelaPC:~/OSSP$ touch bin/my_shell
vennela_07@VennelaPC:~/OSSP$ touch docs/{design.md,report.pdf}
vennela_07@VennelaPC:~/OSSP$ touch tests/{test_parser.c,test_executor.c}
vennela_07@VennelaPC:~/OSSP$ touch scripts/run.sh
chmod +x scripts/run.sh
vennela_07@VennelaPC:~/OSSP$ touch assets/architecture.png
vennela_07@VennelaPC:~/OSSP$ tree
.
├── LICENSE
├── Makefile
├── README.md
├── assets
│   └── architecture.png
├── bin
│   └── my_shell
├── docs
│   ├── design.md
│   └── report.pdf
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── obj
├── scripts
│   └── run.sh
├── src
│   ├── builtin.c
│   ├── executor.c
│   ├── main.c
│   ├── parser.c
│   └── shell.c
└── tests
    ├── test_executor.c
    └── test_parser.c

9 directories, 19 files
```

2. To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing.

```
vennela_07@VennelaPC:~/OSSP$ cd ~/OSSP
mkdir -p process
cd process
nano fork_demo.c
mkdir: cannot create directory 'process': File exists
-bash: cd: process: Not a directory
```

```
vennela_07@VennelaPC: ~/O! × + ▾
GNU nano 7.2 fork_demo.c *
#include <stdio.h>
#include <unistd.h>

int main() {
    pid_t pid;

    pid = fork();

    if (pid == 0) {
        printf("Child Process\n");
        printf("Child PID = %d\n", getpid());
        printf("Parent PID = %d\n", getppid());
    }
    else {
        printf("Parent Process\n");
        printf("Parent PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);
    }

    return 0;
}
```

```
vennela_07@VennelaPC:~/OSSP$ cd ~/OSSP
mkdir -p process
cd process
nano fork_demo.c
mkdir: cannot create directory 'process': File exists
-bash: cd: process: Not a directory
vennela_07@VennelaPC:~/OSSP$ gcc fork_demo.c -o fork_demo
vennela_07@VennelaPC:~/OSSP$ ./fork_demo
Parent Process
Parent PID = 601
Child PID = 602
Child Process
Child PID = 602
Parent PID = 285
```

